

Supplementary Material

Equilibrium Bifurcations and Periodic Dynamics in a Planar Restricted Five-Body Problem with Four Unequal Primaries

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This document collects material supporting the main text: the refinement displacements and full recomputed equilibrium coordinates of the reference cases, the Cartesian branch diagrams, the complete set of local fold panels, the remaining periodic-orbit family figures and candidate ranking, the individual centre frequencies, and a summary of numerical-robustness checks. All tables and figures are generated from the same result files as the main text.

S1 Baseline configurations

The four representative baseline configurations with their computed equilibria (the counts 5, 9, 5, 5 are established in the main text).

S2 Refinement of the baseline configurations

Displacements used to project each configuration onto $\psi = 0$ (holding a, b fixed); the adopted single-coordinate move, the alternative single-coordinate move, and the distance to the constrained nearest CC point (both coordinates free).

S3 Full equilibrium coordinates

Every computed equilibrium of the four baseline configurations, with gradient residual, Hessian determinant, adopted Jacobi value, and linear-stability classification.

S4 Cartesian branch diagrams

The Cartesian branch coordinates $x_i(s)$ and $y_i(s)$ along the two admissible arcs (the radial coordinate $r_i(s)$ is in the main text). They show the same fold and mass-vanishing events.

S5 Local panels for all seven folds

Local structure of every interior saddle-node: the x -coordinates of the merging pair against arclength $\ell - \ell_c$ (three representatives appear in the main text).

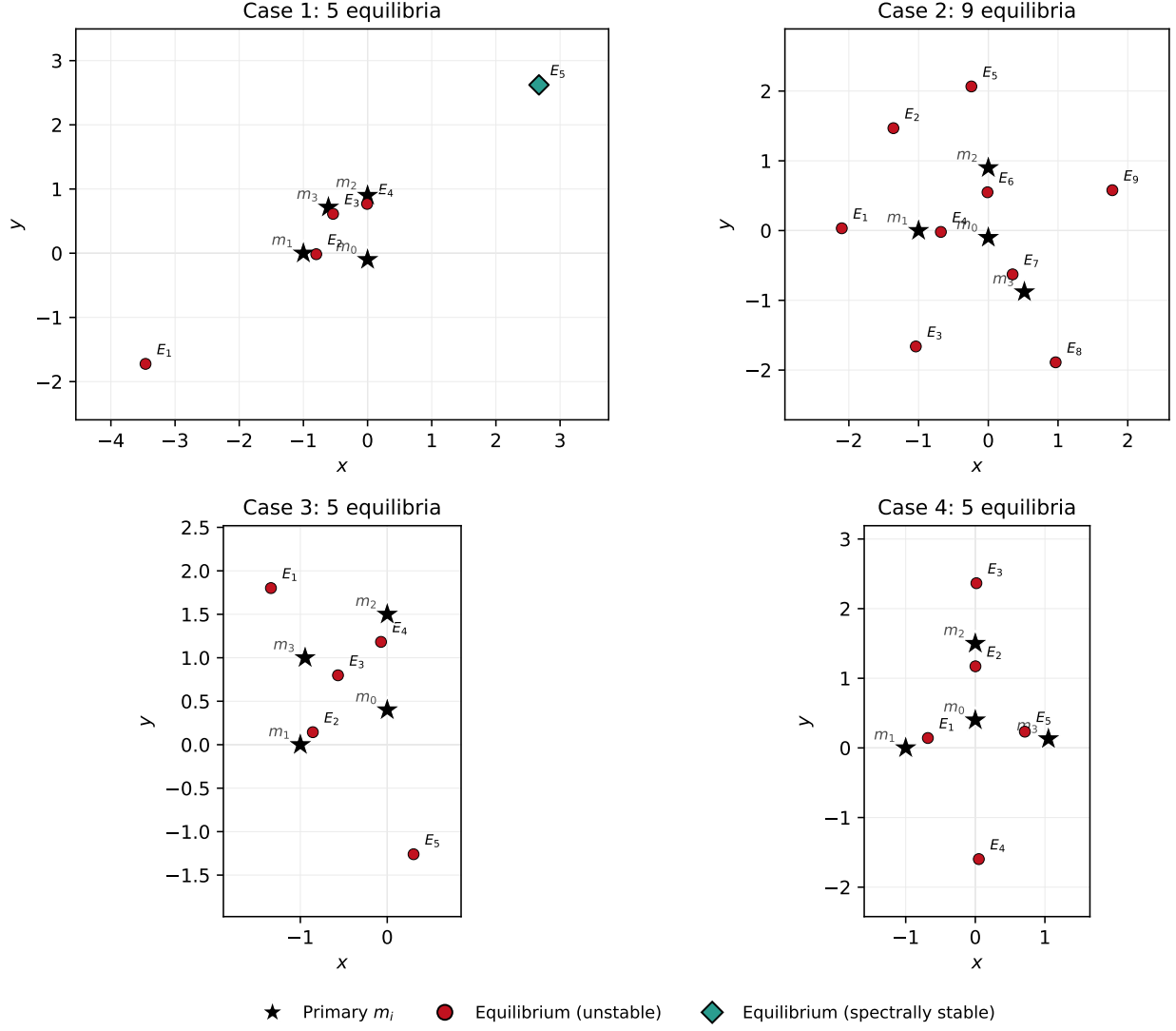


Figure S1: The four baseline configurations. Black stars: primaries at their refined positions; filled circles: equilibria located independently ($\|\nabla\Omega\| < 10^{-12}$); one Case-1 equilibrium near $(2.67, 2.62)$ (green diamond) is spectrally stable.

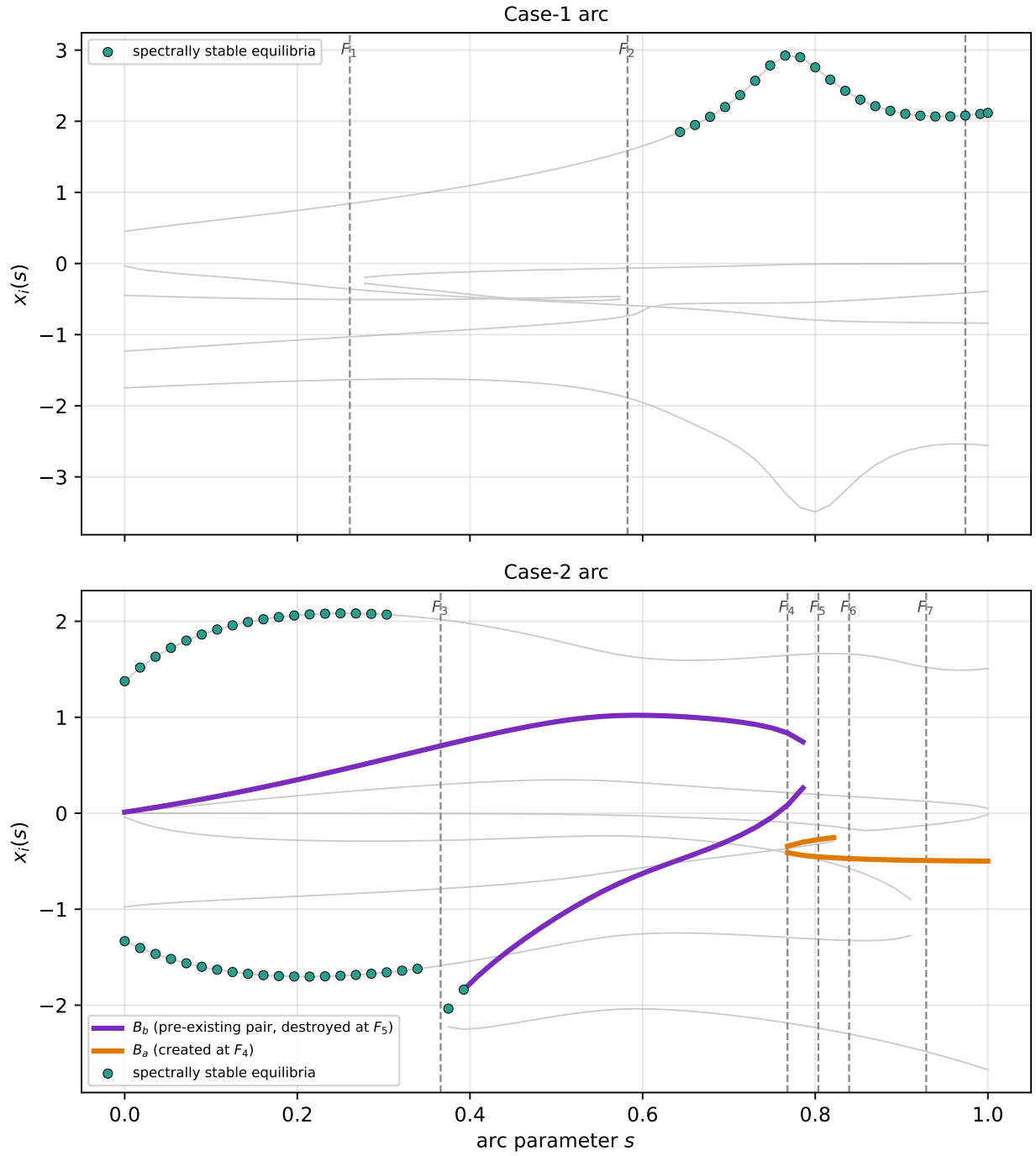


Figure S2: Branch coordinate $x_i(s)$; dashed verticals mark the folds.

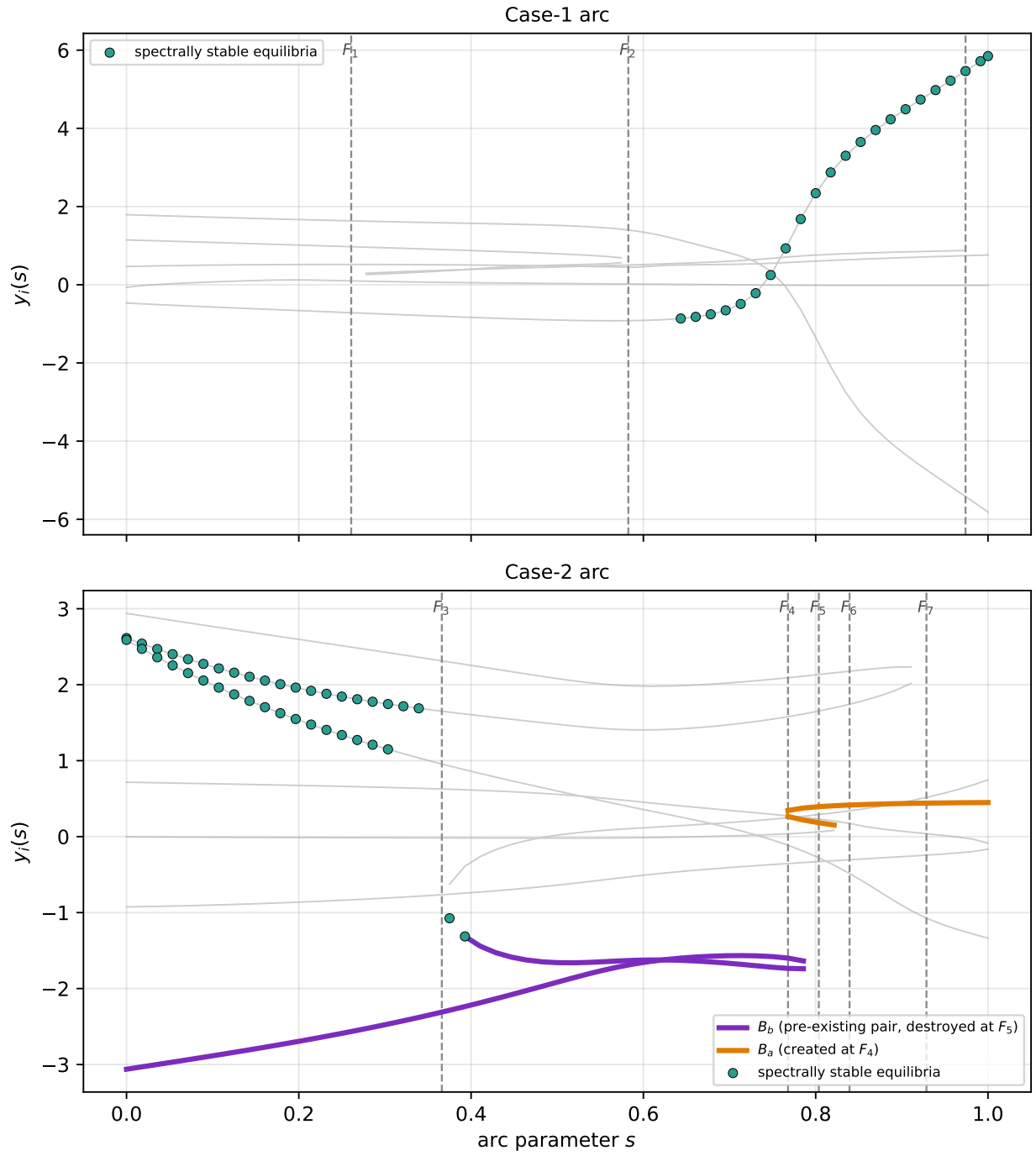


Figure S3: Branch coordinate $y_i(s)$; dashed verticals mark the folds.

Local branch structure at each saddle-node (merging pair coordinates; fold at $\ell = \ell_c$)

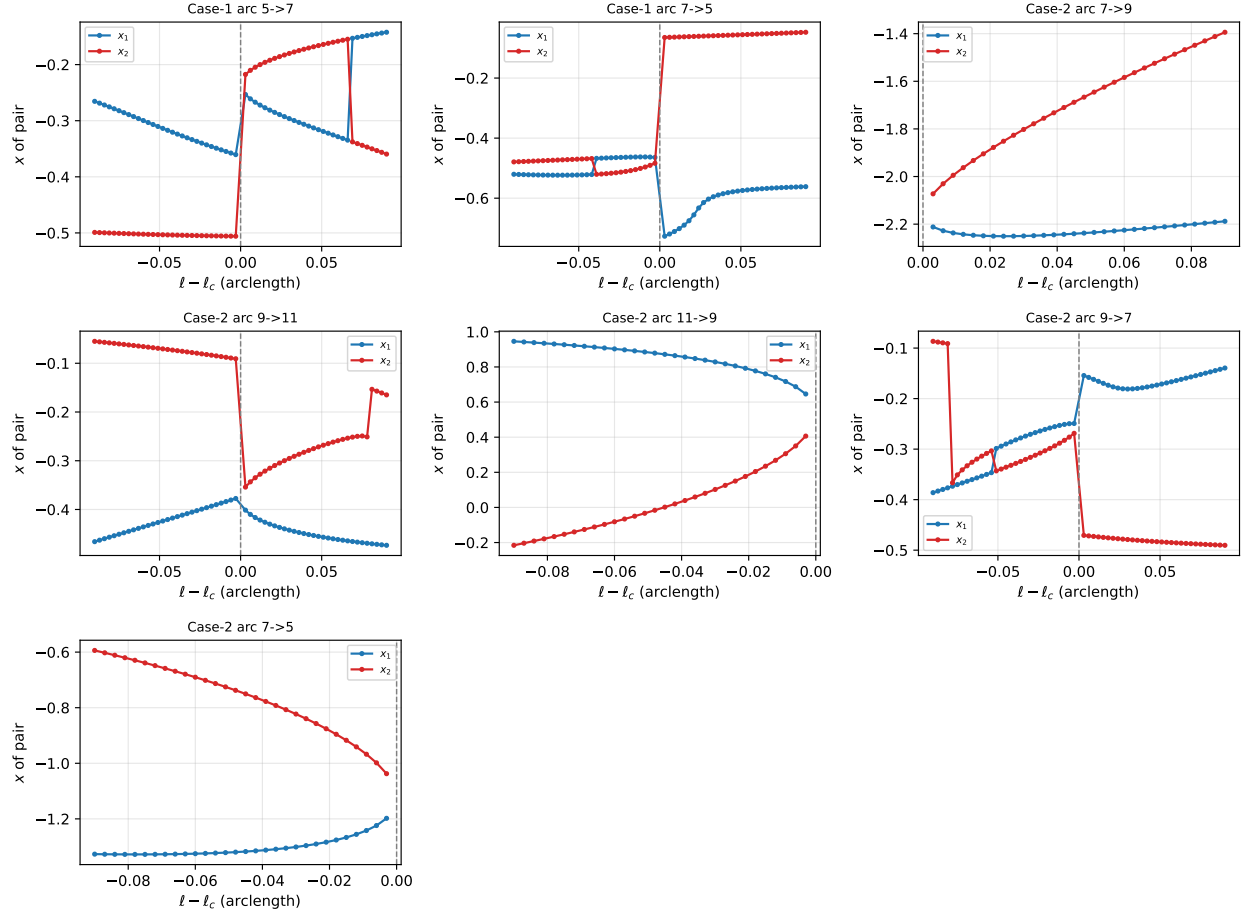


Figure S4: Merging-pair coordinates at all seven folds; the pair annihilates at $\ell = \ell_c$ (dashed).

Table S1: Refinement of each approximate configuration onto $\psi = 0$. The single coordinate with the smaller displacement is adopted; the alternative single-coordinate displacements and the distance to the constrained nearest CC point (both coordinates free) are listed for transparency.

Case	var	Δ	$ \psi _{\text{before}}$	$ \psi _{\text{after}}$	Δc_1^{opt}	Δc_2^{opt}	min-norm
Case 1	$c2$	5.7×10^{-3}	6.0×10^{-3}	8.8×10^{-17}	-1.2×10^{-2}	5.7×10^{-3}	5.1×10^{-3}
Case 2	$c1$	-2.3×10^{-3}	9.2×10^{-3}	9.5×10^{-15}	-2.3×10^{-3}	4.0×10^{-3}	2.0×10^{-3}
Case 3	$c1$	1.5×10^{-2}	7.4×10^{-2}	3.6×10^{-15}	1.5×10^{-2}	2.8×10^{-2}	1.5×10^{-2}
Case 4	$c1$	5.8×10^{-4}	3.4×10^{-2}	3.6×10^{-15}	5.8×10^{-4}	-1.9×10^{-3}	5.6×10^{-4}

S6 Additional periodic-orbit families

The two stable Lyapunov families not shown in the main text.

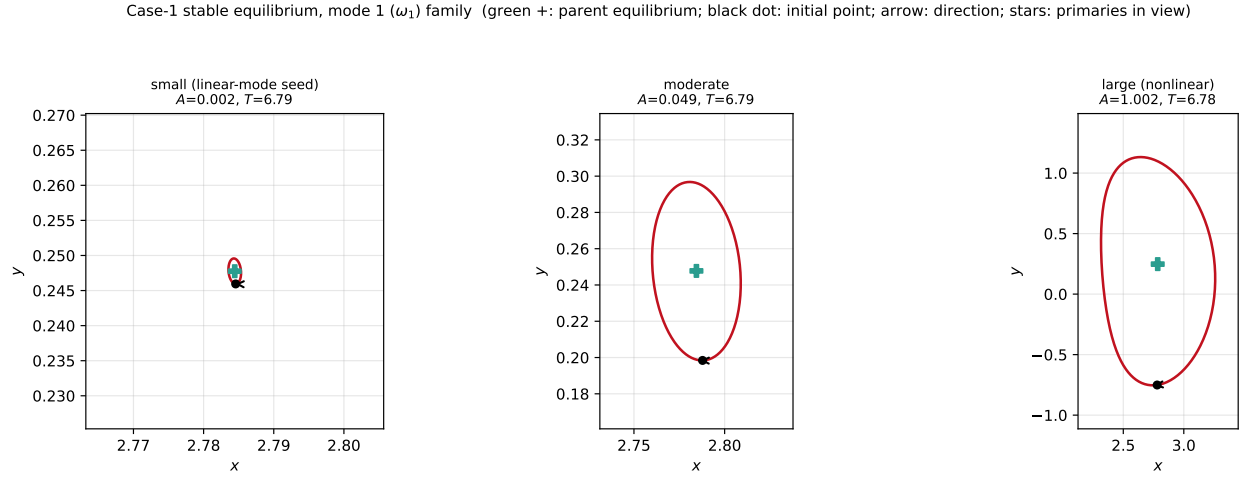


Figure S5: Case-1 arc stable equilibrium, mode-1 (ω_1) family.

S7 Individual centre frequencies

S8 Stable versus unstable orbit on the Hill geometry

S9 Ranked centre-mode candidates

The centre-mode intervals ranked as periodic-orbit continuation candidates (the computed families are summarized in the main text).

S10 Numerical robustness

We summarize the convergence and robustness checks; full numerical settings and file provenance are in the reproducibility protocol.

- **Equilibrium counts vs. search resolution.** Counts are stable under refinement of the global multi-start search. Near a vanishing mass ratio the light-mass equilibria cluster tightly

Table S2: Independently recomputed equilibria at the refined configurations. Each root satisfies $\|\nabla\Omega\| < 10^{-12}$. $D = \det H\Omega = \Omega_{xx}\Omega_{yy} - \Omega_{xy}^2$; $C_J = 2\Omega$ is the adopted Jacobi value at the equilibrium; $\max \text{Re } \lambda$ is taken over the four eigenvalues of the variational matrix. ‘Unstable’ denotes $\max \text{Re } \lambda > 10^{-8}$; ‘spectrally stable’ denotes a purely imaginary spectrum (linear statement only).

Case	i	x	y	$\ \nabla\Omega\ $	D	$\max \text{Re } \lambda$	class
Case 1	1	-3.46033	-1.72387	$< 10^{-16}$	-0.069	0.2587	unstable
Case 1	2	-0.79965	-0.01525	1.3×10^{-14}	-4.5×10^5	30.84	unstable
Case 1	3	-0.53766	0.61300	3.5×10^{-14}	-6.3×10^5	33.73	unstable
Case 1	4	-0.00736	0.76860	4.3×10^{-14}	-6.6×10^5	34.02	unstable
Case 1	5	2.67101	2.62208	$< 10^{-16}$	0.0826	2.0×10^{-15}	spectrally stable
Case 2	1	-2.10151	0.03278	9.5×10^{-16}	-0.934	0.8291	unstable
Case 2	2	-1.36116	1.46813	2.2×10^{-16}	0.6338	0.442	unstable
Case 2	3	-1.03954	-1.66117	2.2×10^{-16}	0.6261	0.415	unstable
Case 2	4	-0.68050	-0.01961	6.3×10^{-15}	-3483	9.132	unstable
Case 2	5	-0.24287	2.06612	2.8×10^{-17}	-0.7812	0.7738	unstable
Case 2	6	-0.01142	0.54921	1.1×10^{-15}	-3976	9.426	unstable
Case 2	7	0.34816	-0.62794	1.2×10^{-15}	-5549	10.22	unstable
Case 2	8	0.96571	-1.88879	4.6×10^{-16}	-1.196	0.9031	unstable
Case 2	9	1.77904	0.57872	2.2×10^{-16}	0.6822	0.409	unstable
Case 3	1	-1.33841	1.80166	$< 10^{-16}$	-3.041	1.323	unstable
Case 3	2	-0.85640	0.14392	2.0×10^{-15}	-812.6	6.483	unstable
Case 3	3	-0.56675	0.79837	1.2×10^{-15}	-824.2	6.385	unstable
Case 3	4	-0.07178	1.18250	2.2×10^{-15}	-516.2	5.794	unstable
Case 3	5	0.30292	-1.25995	2.3×10^{-16}	0.7515	0.3409	unstable
Case 4	1	-0.68158	0.14200	2.3×10^{-15}	-2333	8.207	unstable
Case 4	2	0.00194	1.17180	1.1×10^{-15}	-1742	7.634	unstable
Case 4	3	0.01572	2.36548	$< 10^{-16}$	-3.379	1.32	unstable
Case 4	4	0.05059	-1.59719	2.5×10^{-16}	0.9505	0.442	unstable
Case 4	5	0.71196	0.23320	3.8×10^{-15}	-2222	8.11	unstable

about their primary; a uniform grid alone under-counts there (returning 6 instead of 7 at one Case-2 boundary point), which is corrected by adding a dense local grid around each primary. With that seeding the counts are stable.

- **Independent confirmation of the $N_{\text{eq}} = 11$ window.** Re-solving at higher resolution (grid 121×121 plus 4000 random starts) yields eleven distinct equilibria with pairwise separation 0.10–0.18 and residuals $\sim 10^{-15}$.
- **Fold-location sensitivity.** Each fold is bisection-bracketed to a tight interval in s and then solved from the augmented system to residual $\leq 10^{-11}$; the located critical (x_c, y_c, c_1, c_2) are insensitive to the continuation step at this level.
- **Scaling-exponent fit.** Fitting $d \propto |\ell - \ell_c|^\gamma$ on the innermost points ($|\ell - \ell_c| \leq 0.04$) gives $\gamma \in [0.44, 0.52]$ for all seven folds; widening the window shifts individual fits by $\lesssim 0.1$ due to higher-order curvature but leaves the consistency with $\frac{1}{2}$ intact.
- **Periodic closure vs. integrator tolerance.** With $\text{rtol} = \text{atol} = 10^{-12}$ (DOP853) every accepted orbit satisfies $\|u(T) - u_0\| < 10^{-10}$ (maximum 9.8×10^{-11} over 149 orbits); the trivial Floquet pair is recovered to $\sim 10^{-7}$.

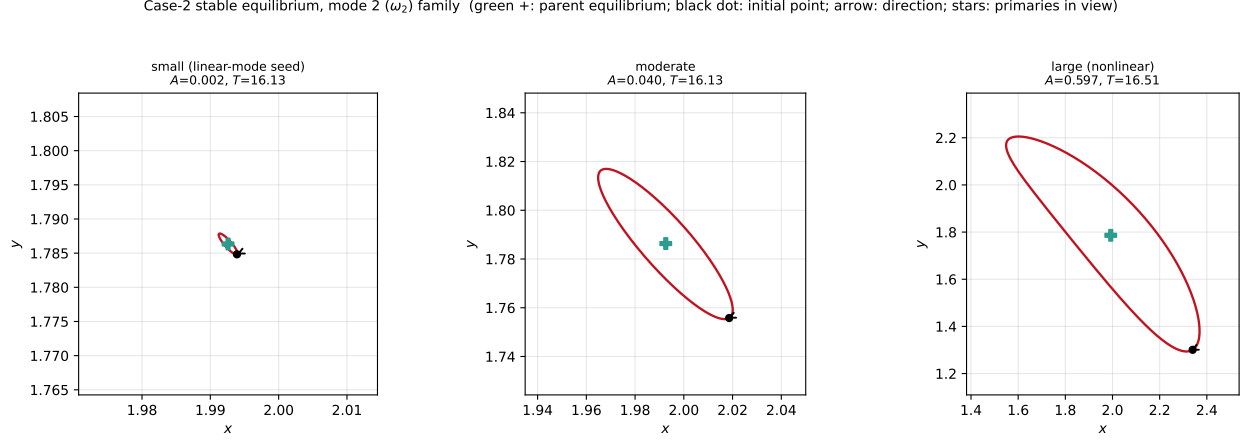


Figure S6: Case-2 arc stable equilibrium, mode-2 (ω_2) family.

Table S3: Top-ranked centre-mode equilibria for future periodic-orbit continuation (ranked by centre-mode interval width, favouring purely-imaginary spectra). ‘centre2’ denotes a spectrally stable (two-centre) interval; ‘saddle $\times$ centre’ a single centre pair. ω_1, ω_2 are centre frequencies at the representative point; $C_J = 2\Omega$. Eigenvectors and continuation parameters (c_1, c_2, s) for each are stored in `results/periodic_orbit_candidates.csv`.

rank	arc / type	s -interval	$(c_1, c_2)_{\text{rep}}$	ω_1	ω_2	C_J
1	Case-1 arc (s $\times$ c)	[0.000, 1.000]	(−0.884, 0.886)	4.284	–	19.855
2	Case-2 arc (s $\times$ c)	[0.000, 1.000]	(0.512, −0.888)	7.435	–	24.294
3	Case-1 arc (s $\times$ c)	[0.000, 1.000]	(−0.884, 0.886)	4.939	–	20.768
4	Case-2 arc (s $\times$ c)	[0.000, 1.000]	(0.512, −0.888)	6.861	–	24.544
5	Case-2 arc (s $\times$ c)	[0.000, 0.911]	(0.487, −0.919)	1.113	–	13.095
6	Case-1 arc (s $\times$ c)	[0.000, 0.835]	(−0.953, 0.960)	1.351	–	11.550
7	Case-2 arc (s $\times$ c)	[0.000, 0.821]	(0.442, −0.959)	8.375	–	29.819
8	Case-2 arc (s $\times$ c)	[0.000, 0.786]	(0.426, −0.970)	1.138	–	15.545

- **Hill connectivity vs. grid.** The one-to-two change in the number of connected components of $\mathcal{H}(C_J)$ at the quoted levels is unchanged at grid resolutions 300, 400, and 500.

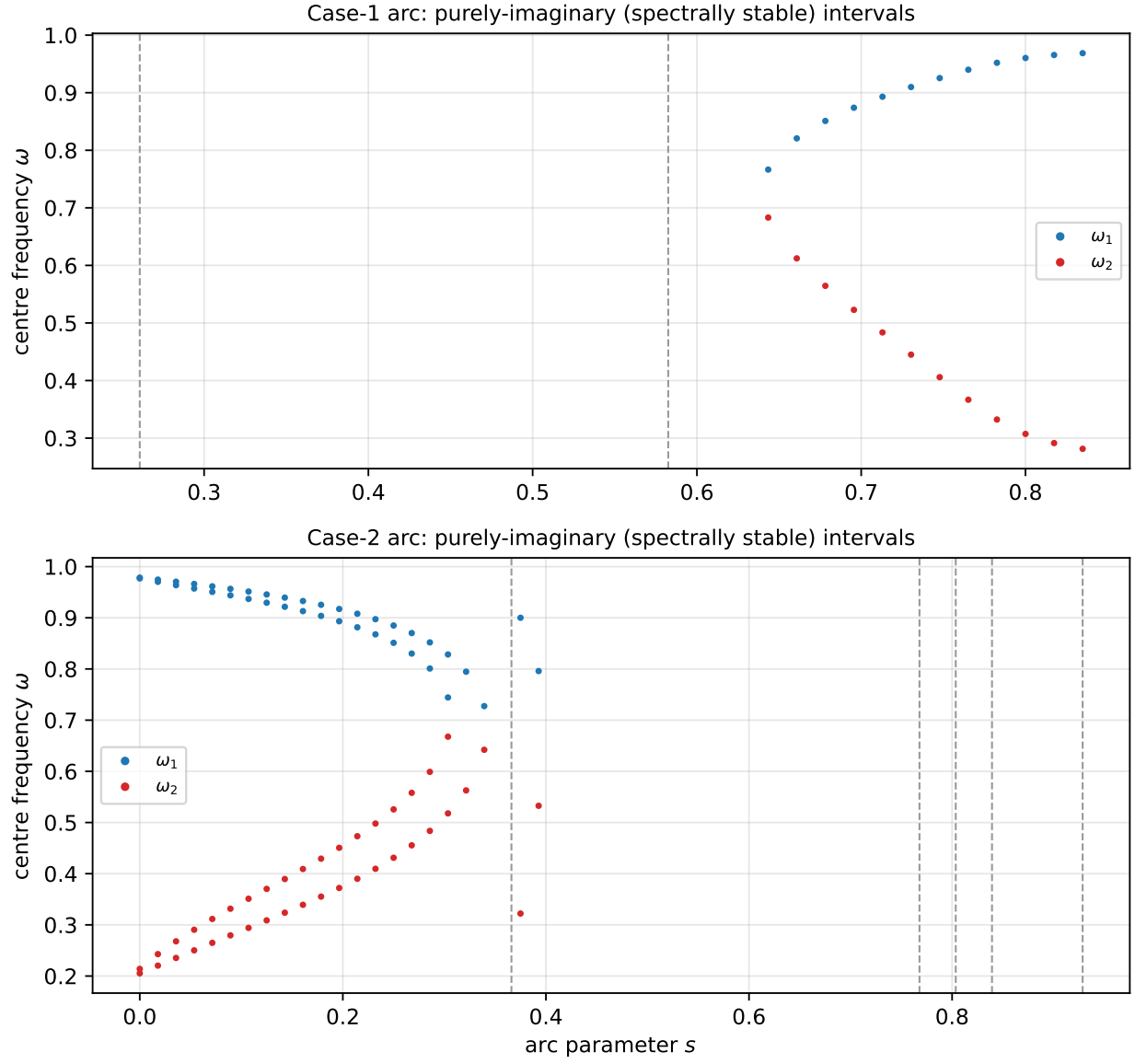


Figure S7: Centre frequencies $\omega_1 \geq \omega_2$ on the spectrally stable intervals of each arc; dashed verticals mark folds.

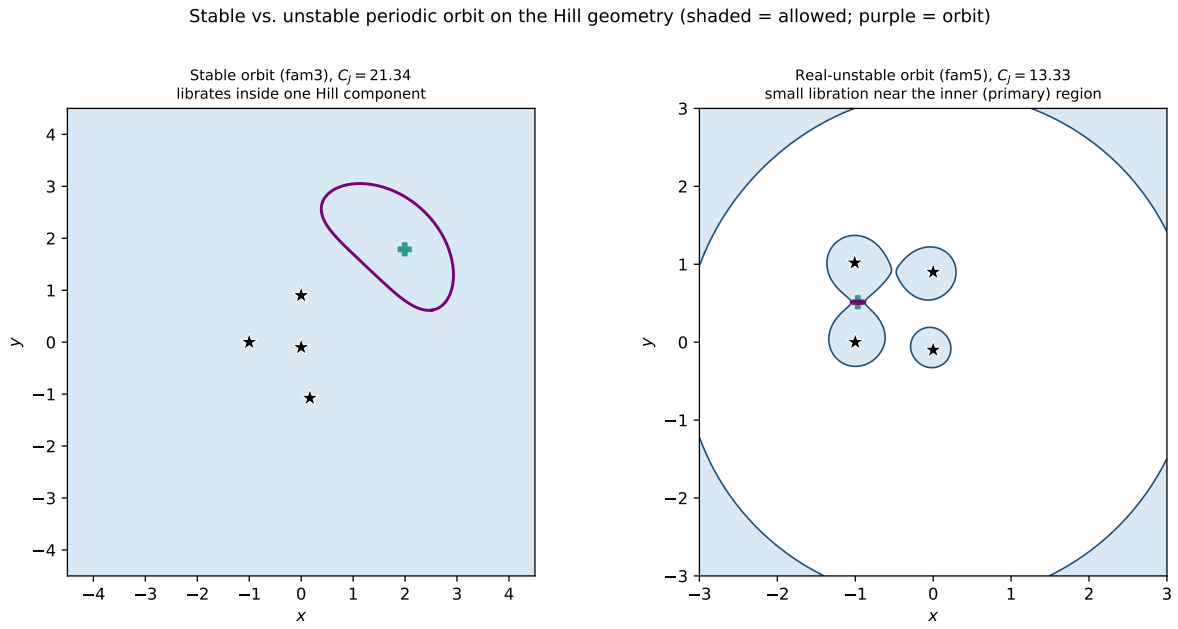


Figure S8: A Floquet-stable orbit (left, fam 3) and a real-unstable orbit (right, fam 5) on their respective Hill geometries at their own C_J (shaded: allowed; blue: zero-velocity curve; purple: orbit; stars: primaries; green +: parent equilibrium).